

PART A: OBJECTIVE QUESTIONS (20 MARKS)

1. Which of the following statements is the specification of **non-ideal pipelining**?
 - A. It has just two segments and only one process can be run in a time. ✗
 - B. It has several segments and all the segments have the same execution time. ✗
 - ☒ C. It consists of several segments with **various execution times**.
 - D. The interface delay for the pipelining is very short.
2. Which of the following statements is **NOT** a limitation in pipelining architecture?
 - A. Resource Conflict ✓
 - ☒ C. Execution Cycle
 - B. Data Dependency ✓
 - D. Branch Difficulties ✓
3. Which of the following is **NOT** one of the **control unit inputs**?
 - A. Instruction register ✓
 - ☒ C. Control signal to memory
 - B. Control signal from control bus ✓
 - D. Flags ✓
4. Micro-operations include the data transferring
 - A. Into or out of a register
 - C. Between register to external bus
 - B. Between register to register ✓
 - ☒ D. All of the above.
5. Which of the following facts about **LOI** or **HOI** is **FALSE**?

Memory Interleaving

 - A. In high-order interleaving (HOI), the high order bits specify the **memory bank**. ✓
 - B. In LOI, a **failure of any single bank** would be **catastrophic to the whole system**. ✓
 - ☒ C. LOI provides better reliability, since a failed module affects only a localized area of the address space. *data distributed across the banks.*
 - D. Easier to extend memory modules in HOI.
6. In **Set Associate Mapping**, read or write a word at the memory address **357A** is equal to 0011 0101 0111 1010. This address will be translated to _____.
 - A. tag = 001101, slot=0101, word = 1111010 ✗
 - B. tag = 001101, line = 01011, word = 11010 ✗
 - C. set=0011, tag = 010101, word = 1111010
 - ☒ D. tag = 001101, set= 010111, word = 1010

Tag set Word

7. Computer memory is organized in a hierarchy, with the _____ i _____, _____ ii _____ memory at the top and the largest, _____ iii _____ memory at the bottom.
- A. i=largest, ii=fastest, iii=slowest
 B. i=largest, ii=slowest, iii=fastest
☒ C. i=smallest, ii=fastest, iii=slowest
 D. i=smallest, ii=slowest, iii=fastest
8. _____ enhances performance by providing greater memory capacity, without the expense of adding main memory. While, _____ memory enhances performance by providing faster memory access speed.
- A. replacement algorithm, virtual
 B. ☒ Virtual memory, cache
 C. Cache memory, virtual
 D. Dynamic memory, static
9. The performance of a replacement algorithm for cache also known as the Average Memory Access Time (AMAT), can be computed using _____.
- A. $AMAT = \text{Hit rate} \times \text{Hit time} + \text{Miss penalty}$
 B. $AMAT = \text{Hit rate} + (\text{Miss rate} + \text{Miss penalty})$
☒ C. $AMAT = \text{Hit time} + (\text{Miss rate} \times \text{Miss penalty})$
 D. $AMAT = \text{Hit ratio} + (\text{Miss rate} \times \text{Miss cycle})$
10. Which of the following is NOT a measure for computer performance?
- A. Response time
 B. RAM size
☒ C. Throughput
 D. Execution time
11. Performance of computer can be measured based on the following EXCEPT?
- A. Speed
☒ B. Throughput
 C. Read Time
 D. Response Time
12. To achieve better performance, RISC machines reduce _____, meanwhile CISC reduce _____.
- A. the number of cycles per instruction, the number of instructions per program
☒ B. the number of instructions per program, the number of cycles per instruction
 C. the program execution time, the clock cycle
 D. the clock cycle, the program execution time

13. Which of these is a type of I/O devices?
- A. Integer I/O devices
☒ B. Block I/O devices
 C. Bit I/O devices
 D. Stream I/O devices
14. The input output subsystems include
- I. Blocks of main memory
 II. Programmed polling
 III. Control module
 IV. Interfaces
 V. Cabling
- A. All of the above
☒ C. I, II, IV and V
 B. I, II, III and IV
 D. I, III and V
15. The input output can be controlled using the input output channel. The I/O processors (IOPs) serves as
- I. Negotiates protocols
 II. Issues device commands
 III. Translates storage coding to memory coding
 IV. Transfer entire files or groups of files independent of the host CPU.
- ☒ A. All of the above
 C. I, II and IV
 B. II, III and IV
 D. III and IV
16. If **AX = 3254H** and **BX = 2B2CH**, what is the result of **AND AX, BX**?
- A. AX = 2202H
 B. AX = 4404H
☒ C. AX = 2204H
 D. AX = 4402H
17. Study the assembly code given. What is the final result of register CH?

```

MOV AH, 3
MOV BH, 14
MOV CH, 7
ADD AH, BH

```

AH = 17

```

JNZ HERE

MOV CH, 1

JMP QUIT

HERE:

MOV CH, 5

QUIT:

```

A. 1
☒ B. 5

C. 7
D. 0

18. Based on the assembly code below. Please identify if the line indicated is a valid code.

```

.DATA

SUM WORD 126

TOTAL WORD 55H


.CODE

MAIN PROC
    16bit 8bit
    MOV AX, BH ; LINE1
    ADD SUM, TOTAL ; LINE2
    SUB AX, SUM ; LINE3

```

mov AX, SUM
add AX, TOTAL
mov SUM, AX (✓)

✗
Memory to memory ✗

A. LINE1 = valid; LINE2 = invalid; LINE3 = valid
☒ B. LINE1 = invalid; LINE2 = invalid; LINE3 = valid
C. LINE1 = invalid; LINE2 = invalid; LINE3 = invalid
D. LINE1 = valid; LINE2 = valid; LINE3 = valid

19. Study the assembly code given below. If the first user input is 12 and the second is 8, what is the correct output?

```
.DATA
STR1 BYTE "ENTER A NUMBER :",0

.CODE

MAIN PROC

MOV EDX, OFFSET STR1
CALL WRITESTRING

CALL READDEC 12

MOV EDX, OFFSET STR1
CALL WRITESTRING

CALL READDEC 8

CALL WRITEDEC

CALL WRITEDEC
```

- A. 12 8
B. 12 12
C. 8 12
D. 8 8
20. Study the assembly code given below. Choose the correct output.

```
.DATA
STR1 BYTE "Test1 ",0
STR2 BYTE "Test2 ",0
TREST BYTE "aBc ", 0
STR3 BYTE "Test3 ",0

.CODE

MAIN PROC

MOV EDX,OFFSET TREST
```

abc

```
MOV EAX, OFFSET STR3
CALL WRITESTRING
CALL CRLF
```

```
MOV EDX, OFFSET STR1
CALL WRITESTRING
```

```
MOV EDX, OFFSET STR2
CALL WRITESTRING
CALL CRLF
```

Test 3

abc

Test1

Test2

A.

```
C:\Windows\system32\cmd.exe
Test3
Test1 Test2
Press any key to continue . . .
```

C.

```
C:\Windows\system32\cmd.exe
Test1
Test2
Test3
Press any key to continue . . .
```

B.

```
C:\Windows\system32\cmd.exe
aBc
Test2 Test3
Press any key to continue . . .
```

D.

```
C:\Windows\system32\cmd.exe
aBc
Test1 Test2
Press any key to continue . . .
```

PART B: STRUCTURED QUESTIONS (80 MARKS)

QUESTION 1 (15 MARKS)

- (a) A pipeline processor has 6 segments - *Fetch Instruction* (FI), *Decode Instruction* (DI), *Calculate Operand* (CO), *Fetch Operand* (FO), *Execute Instruction* (EI) and *Store Operand* (SO). There are five instructions (I1 – I5) involved, they are:

I1: CMP AH, AL

I2: MUL CH, 2

I3: JMP I5

I4: ADD DH, 011F

I5: DIV BX, 3

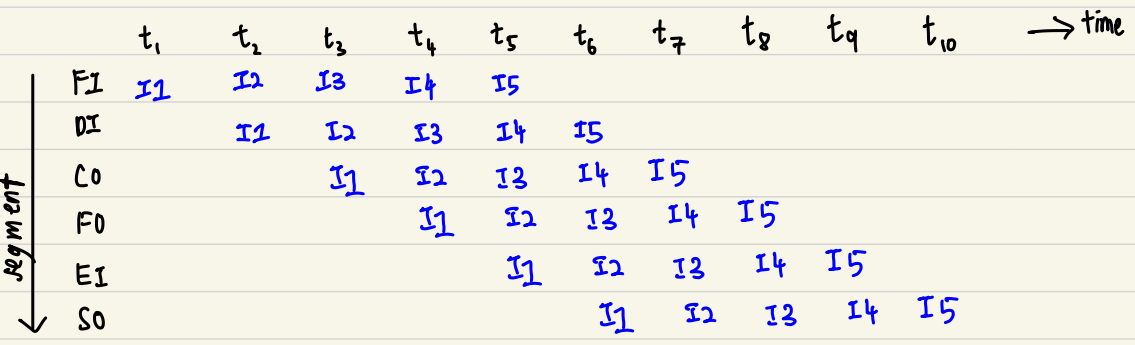
- Draw the **space time diagram** for the above instructions. [3]
 - Solve the branching problem using **NOP**. [4]
- (b) Table 1 shows the delay time for four segments in a pipelining architecture. If the **interface delay is 10 nanoseconds**, calculate:

Table 1: Delay time for segments in pipelining.

Segment Name	Delay Time (nanoseconds)
Segment 1	40
Segment 2	25
Segment 3	30
Segment 4	30

- The cycle time of the pipeline, t_p . [2]
- The execution time for 200 tasks. [4]
- The real and the maximum speed up for 200 tasks. [2]

1.(a) (i) Space Time Diagram



(ii) 3-1
no. of NOP = 2



$$(b)(i) \quad t_p = 40 + 10 \\ = 50 \text{ ns}$$

$$(iii) \quad S_{max} = k \\ = 4$$

$$(ii) \quad T_p = t_p (k + (n-1)) \\ = 50 (4 + (200-1)) \\ = 10150$$

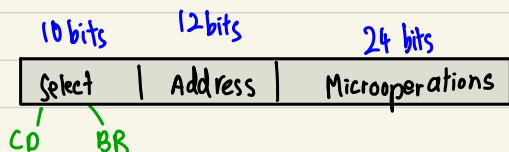
Real speed up.

$$S = \frac{T_n}{T_p}$$

$$T_n = t_n \times n \\ = (40 + 25 + 30 + 30 + 10) \times 200 \\ = 27000 \text{ ns}$$

$$= \frac{27000}{10150} \\ = 2.66$$

2. (a)



$$4096 \text{ words} = 2^{12}$$

address field : 12 bit

$$\text{selection field} : 46 - 24 - 12 \\ = 10 \text{ bit}$$

(b) 64 status means 6 bits are required for it (2^6).

/
 CD

$$(c) \text{ Input for the } = \text{selection field bits} - \text{status bits} \\ \text{multiplexer} = 10 - 6 \\ = 4 \text{ bits}$$

BR

3. (a) SRAM - used in cache memory & registers where speed is crucial.
- more expensive & consume more power.

DRAM - used in main memory (RAM)
- less expensive, allowing for more memory in a smaller space.

(b) No. of locations = 2^{24}

(i) = 16777216 addresses

$$\begin{aligned}\text{Memory capacity} &= 16777216 \times 24 \text{ bits} \\ &= 402653184 \text{ bits} \div 8 \text{ bits per byte} \\ &= 50331648 \text{ bytes} \div 2^{20} \\ &= 48 \text{ MB}\end{aligned}$$

(ii) No. of block = $\frac{\text{Total memory}}{\text{block size}}$ each block = 64 Kbit

$$\begin{aligned}&= \frac{32 \text{ MB}}{64 \text{ Kbit}} \\ &= \frac{32 \times 2^{20} \times 8 \text{ bit}}{64 \times 2^{10}} \\ &= 4096 \text{ block}\end{aligned}$$

(iii) Page size = $\frac{\text{Total memory}}{\text{Number of pages}}$

$$\begin{aligned}&= \frac{2 \text{ MB}}{128} = \frac{2 \times 2^{20}}{128} = 16384 \text{ Byte} \div 2^{10} \\ &= 16 \text{ KB}\end{aligned}$$

QUESTION 2 (5 MARKS)

1/2 word 46 bits

A system uses a control memory of 4096 words of 46 bits each. The microinstruction has three fields, i.e., select, address and microoperations fields.

- (a) If the microoperation field requires 24 bits, how many bits are there in the branch address field and the select field? [2]
- (b) If there are 64 status in the system, how many bits of the branch logic are used to select a status? [2]
- (c) How many bits are left to select an input for the multiplexers? [1]

QUESTION 3 (20 MARKS)

- (a) Differentiate between SRAM and DRAM. [2]
- (b) Based on the following specifications: [6]
 - i. 24-bit memory address, each address can hold 24 bits. What is the memory capacity?
 - ii. 32 Mbytes memory divided into 64-kilobit blocks. How many blocks available?
 - iii. 2 Mbytes memory is divided into several pages. There are 128 pages. What is the size of each page?
- (c) Given a memory address as 34Ah (10 bits) and there are 4 memory banks. Determine the memory bank address and the address of the word in the bank using LOI. [4]
- (d) A computer main memory is utilizing a Block Direct Mapping:
 - i. Given the main memory address format as below, what is the total memory capacity, if each main memory word is 8 bits? [3]

5	8	3
Tag	Line	Word

- ii. The total numbers of cache words? [2]
- iii. Suppose a program generates an address 246Ah and then the program generates the address 256Ah, will it be a hit or a miss? Show your working. [3]

QUESTION 4 (10 MARKS)

(a) Identify the correct terms for **A to E**. Select from answer options below. [5]

Description	Term
It consists one or more I/O processors (IOPs) that control various channel paths.	i Channel I/O
It is less wasteful of CPU cycles than programmed I/O.	ii Interrupt Driven I/O
It has direct access to the memory (steals bus cycles from the CPU).	iii Direct Memory Address (DMA)
It delegates I/O to intelligent I/O device controllers called I/O processors that can execute programs with branching instructions.	iv Programmed I/O
Two dominant recording methods are serpentine and helical scan, which are distinguished by how the read-write head passes over the recording medium.	v Magnetic Tape

Answer options:

Channel I/O	Interrupt Driven I/O	Programmed I/O
Direct Memory Address	Magnetic Tape	Optical Disks

(b) The following parameters are used for processor overhead calculation for interrupt I/O and DMA. [5]

- 500 MHz CPU
- Interrupt handler takes 400 cycles
- Data transfer takes 100 cycles
- 4 MB/s, 16 B interface disk transfers data 50% of time
- DMA setup takes 800 cycles, transfer one 16KB page at a time

- Calculate processor overhead for interrupt I/O.
- Calculate processor overhead for DMA.

QUESTION 5 (20 MARKS)

(a) Given the pseudocode below, find the arithmetic expression for it. [4]

MOV AX, SUMA	$AX = SUMA$
MOV BL, SUMB	$BL = SUMB$
DIV BL	$AL = \frac{AX}{BL}$
INC AL	$AL = \frac{SUMA}{SUMB} + 1$
MOV TOTAL, AL	$TOTAL = AL$

$AL = \left(\frac{SUMA}{SUMB} \right) + 1$

(b) Given the pseudocode below, find the content of registers as indicated. [3]

MOV CL, BFh	$11\ 15$	8421	$= 1011\ 1111$
MOV BL, 10h			$= 0001\ 0000$
NOT CL			$; CL = 0100\ 0000 = 40h$
AND BL, CL			$; BL = 0001\ 0000 = 10h$
OR BL, 5			$; BL = 0001\ 0101 = 15h$

Final BL value: $0001\ 0101 = 15h$

- (c) Given the array definition below, write the complete assembly code to sum the content of array that in odd position by using *loop*. Then, print the value of sum (in decimal) as an output. [7]

```
.data
Array2 word 25h, 1A20h, 66h, 891h, Ah
```

- (d) Given the pseudocode below, write the appropriate assembly code to achieve it. [6]

```
If BX < AX AND Y = 0 THEN
Y = Y - 1
ELSE
Y = BX + Y
```

QUESTION 6 (10 MARKS)

- (a) Table 2 depicts the CPU Execution Time of Computer A and B.

Table 2: CPU Execution Time for Computer A and B

	Clock cycle	CPI	Clock rate
Computer X	1000ps	3.0	4GHz
Computer Y	1500ps	2.0	4GHz

Note: The formula

CPU time = Instruction count x CPI x Clock cycle time

Or

CPU time = $\frac{\text{Instruction count} \times \text{CPI}}{\text{Clock rate}}$

(c) .data

Array2 word 25h, 1A20h, 66h, 891h, 0Ah
sum dword ?

.code

main PROC

mov esi, OFFSET Array2
mov ecx, LENGTHOF Array2 / 2
xor eax, eax

mov edi, 1

sum_loop:

add ax, word ptr [esi + edi * 2]
add edi, 2
loop sum_loop

mov sum, eax

mov eax, sum
call WriteDec
call Crlf

exit

main ENDP

END main

(d) TITLE

INCLUDE Irvine32.inc

.data

Y DWORD 0

.code

main PROC

mov ax, 5

mov bx, 3

mov eax, Y

cmp bx, ax

jge else_part ; jge (jump if greater or equal)

cmp eax, 0

jne else_part ; jne (jump if not equal)

dec eax

jmp END_IF

else_part:

add eax, ebx

END_IF:

mov Y, eax

exit

main ENDP

END main

- i. Calculate the CPU time for each computer. [4]

$$\begin{aligned} X &= 1000 * 3.0 * 4 * 10^9 \\ &= 12000 * 10^9 \\ &= 1.2 * 10^{9+4} \\ &= 1.2 * 10^{13} \end{aligned}$$

G

$$\begin{aligned} Y &= 1500 * 3.0 * 4 * 10^9 \\ &= 18000 * 10^9 \\ &= 1.8 * 10^{9+4} \\ &= 1.8 * 10^{13} \end{aligned}$$

- ii. From the CPU time calculated, which computer should be selected based on the performance? Discuss your observation. [3]

X, execution time < Y

- (b) Explain the important of benchmark in CPU design. [3]

identify the systems that provide good performance at the most reasonable cost.

Benchmark provide a standard measure to compare the performance of different CPUs.

- This help designers and users understand which CPU performs better under specific workloads.